



BENGALURU CITY UNIVERSITY

CHOICE BASED CREDIT SYSTEM

(as per SEP 2024)

Syllabus for I & II Semester B.Sc. Botany

2024-25

COURSE PATTERN AND SCHEME OF EXAMINATION FOR B.SC. DEGREE AS PER SEP (2024-25 ONWARDS)																
SUBJECT : BOTANY																
Sl. No	Seme ster	Title Of The Paper	Teachi ng Hours	Hours/Week		Examination pattern Max. and Min. Marks/Paper						Duration of exam (hours)		Total mark s/ paper y	Credits	
				Theory	Practic al	Theory			Practical			Theory	Practical		Theor y	Pract ical
						Max.	Min.	IA	Max.	Min.	IA					
1	I	CORE SUBJECT	60	4	3	80	32	20	40	16	10	3	3	150	3	2
2	II	CORE SUBJECT	60	4	3	80	32	20	40	16	10	3	3	150	3	2

Proceedings of the meeting of BoS (UG) in Botany held on 5th & 6th July 2024
at the Department of BCU School of Management, Jnana Jyothi Campus,
Bangalore City University, Bengaluru – 560 001

Venue: Department of BCU School of Management, Jnana Jyothi
Campus, Bangalore City University, Bengaluru – 560 001

Date: 06/07/2024

Time: 11:00 AM

Agenda:

1. To finalize the syllabus for I and II Semester B.Sc. Botany (UG) (CBCS) SEP-2024 for approval.
2. To approve the panel of examiners recommended for the examinations of 2024-25.
3. To recommend and approve the constitution of BoE for the academic year 2024-25.

Members Present

Signature

1. Smt. Zaiba Nishath Bano

Member

Zaiba Nishath Bano

2. Dr. Mallikarjuna P.B.

Member

Dr. Mallikarjuna P.B.

3. Dr. B. L. Manjula

Member

Dr. B. L. Manjula

4. Smt. Shobharani

Co-opted Member

Smt. Shobharani

5. Esther Watson

Co-opted Member

Esther Watson

6. Roopashree M. G.

Co-opted Member

Roopashree M. G.

7. Dr. N. S. Suresha

Co-opted Member

Dr. N. S. Suresha

8. Dr. L. Rajanna

CHAIRMAN

Dr. L. Rajanna
06/7/24

Members Absent

1. Smt. K. R. Kavitha

Member

2. Smt. Chandrakala Shivakumar

Member

MINUTES OF THE MEETING OF BoS (UG) IN BOTANY

Chairman welcomed the members of the BoS (UG) in Botany to the meeting and the agenda was placed for discussion.

- a). Discussed and finalized the syllabus for theory and practical of I and II Semester B.Sc., Botany (CBCS) SEP 2024, question paper pattern, blue print of question paper Formative assessment and Scheme of valuation for SEP programme to be implemented from the academic year 2024-25.
- b). The panel of Examiners was approved and recommended for UG Examination for the academic year 2024-25.
- c). Recommendations were made to constitute BoE for the academic year 2024-25.
- d). The Chairman was authorized to change / incorporate the corrections as per the directions of Bangalore City University.
- e). Discussed about the III, IV, V & VI semester papers to be introduced

The meeting ended with a vote of thanks by the Chairman.

1. Smt. Zaiba Nishath Bano *Zaiba*
2. Dr. Mallikarjuna P.B. *Mallikarjuna*
3. Dr. B. L. Manjula *Manjula (STEC, Race Course Road, Bellur-1)*
4. Smt. Shobharani *Shobharani (GFGC Yelahanka)*
5. Esther Watson *Esther (Bishop Cotton Women's Christian College)*
6. Roopashree M. G. *Roopashree (KLE SOCIETY'S Nijalingappa College)*
7. Dr. N. S. Suresha *N. S. Suresha*
8. Dr. L. Rajanna *L. Rajanna*

Dr. L. RAJANNA
Senior Professor
Department of Botany
Bangalore University
BANGALORE - 560 056.

BOTANY CURRICULUM

I SEMESTER

PAPER – I: MICROBIAL DIVERSITY, MYCOLOGY AND PHYCOLOGY

Programme name	B. Sc. BOTANY	SEMESTER	I
Course Title	PAPER – I: MICROBIAL DIVERSITY, MYCOLOGY AND PHYCOLOGY		
Course Code	BOTT - 101	No. of Credits	03
Contact Hours	60 Hours	Duration of Exam	03 Hours
Formative Assessment Marks	20	Summative Assessment Marks	80

Course objectives:

- To understand the microbial diversity through isolation techniques from various environments, mastering methods of sterilization and learning microbial culture and preservation techniques.
- Students will explore the structure, classification and multiplication of viruses like TMV and Bacteriophage T4, along with the economic importance of viruses and vaccination strategies
- Students will explore the structure, reproduction of Fungi, economic importance and disease caused by Fungi.
- The course also covers bacterial characteristics, reproduction, plasmid biology and bacterial diseases such as Citrus canker and Phytoplasma – related diseases.
- Additionally, students will study Cyanobacteria, algae(Phycology) and their economic roles and the environmental applications of algae in industries and agriculture.

Course outcomes:

- To understand the fascinating diversity, evolution and significance of microorganisms.
- To comprehend the systematic position, structure, physiology and life cycles of microbes and their impact on humans and environment.
- To gain laboratory skills such as microscopy, microbial cultures, staining, identification, preservation of microbes and their applications in research and industries.

- To understand the diversity and affinities among Fungi, Cyanobacteria and Algae and their applications in Biotechnology and industries.

I SEMESTER

PAPER – I: MICROBIAL DIVERSITY, MYCOLOGY AND PHYCOLOGY

60 Hrs.

UNIT I	Microbiology - Introduction and Scope of Microbiology. Branches of microbiology - Industrial, Medical, Agricultural and Environmental Microbiology. Contributions of scientists to the field of Microbiology -Anton von Leeuwenhoek, Louis Pasteur, Robert Koch and Iwanowsky. Microbial culture techniques -Culture media, serial dilution and pour plate method. Viruses –Properties, structure of TMV & T4 Bacteriophage, Multiplication (lytic and lysogenic cycle) & Transmission of viruses. Tomato leaf curl disease. Brief account of Prions and Viroids	15 Hrs.
UNIT II	Bacteriology - Introduction, classification of Bacteria based on shape and flagella. Ultra structure of bacterial cell with special reference to Gram positive and Gram negative cell wall composition. Endospore – a brief account. Reproduction – Binary fission and genetic recombination (Transformation, Transduction and Conjugation). A brief account of plasmids – definition and properties. Structure and importance of Ti plasmid. Bacterial nutrition, Citrus canker disease Economic importance – Role of bacteria in Agriculture, Medicine and Industry General account of Phytoplasma and Sandal spike disease.	15 Hrs.
UNIT III	Mycology - General characters and reproduction in fungi. Structure, reproduction and life cycle of <i>Albugo</i>, <i>Peziza</i> and <i>Puccinia</i>. Economic importance - Role of fungi in Medicine, Agriculture and Industry. Plant Diseases -Tikka disease of Groundnut, Red rot of Sugarcane, Grain smut of Sorghum and Koleroga of Arecanut. Lichens - General account, reproduction and ecological importance. A brief account of Mycorrhiza and Biopesticides	15 Hrs.
UNIT IV	Cyanobacteria and Phycology Cyanobacteria: Introduction, general characteristics, reproduction, and economic importance. Type study: <i>Anabaena</i> and <i>Scytonema</i>. Algae - General characters, occurrence, structure, reproduction and life cycle of <i>Chlamydomonas</i>, <i>Hydrodictyon</i>, <i>Spirogyra</i>, <i>Chara</i>, Diatoms, <i>Sargassum</i>, and <i>Polysiphonia</i>. Economic importance of Algae in Agriculture, Medicine and Industry	15 Hrs.

Assessment:	Marks:
Attendance	05 Marks
Assignment	05 Marks
Test	10 Marks
Total	20 Marks

PRACTICAL PAPER – I

PAPER – I: MICROBIAL DIVERSITY, MYCOLOGY AND PHYCOLOGY

No of Credits: 02

45 Hrs.

	BOTP – 102; MICROBIAL DIVERSITY, MYCOLOGY AND PHYCOLOGY	15 Units
1.	Study of instruments: autoclave, inoculation chamber/LAF, hot air oven, incubator and inoculation loop.	1 Unit
2	Sterilization of glassware, media preparation - nutrient media (Nutrient Agar), Isolation of Bacteria from soil by pour plate method, Colony characteristics of Bacteria to identify colonies.	1 Unit
3	Diseases: Citrus Canker, Sandal spike, Tomato leaf curl, Tikka disease of Groundnut, Red rot of Sugarcane, Grain smut of Sorghum and Koleroga of Arecanut.	2 Units
4	Gram staining: Rhizobium from root nodules and Lactobacillus from curd.	1 Unit
5	Measurement of cell concentration – yeast cells / fungal spores using Haemocytometer	1 Unit
6	Type study of Cyanobacteria: <i>Anabaena</i> and <i>Scytonema</i>	1 Unit
7	Type study of Algae: <i>Chlamydomonas</i> , <i>Hydrodictyon</i> , <i>Spirogyra</i> , <i>Chara</i> , <i>Diatoms</i> , <i>Sargassum</i> and <i>Polysiphonia</i>	5 Units
8	Type study of fungi: <i>Albugo</i> , <i>Peziza</i> and <i>Puccinia</i>	2 Units
9	Lichens and Mycorrhiza	1 Unit

Assessment:	Marks:
Continuous assessment/Attendance	05 Marks
Test	05 Marks
Total	10 Marks

PRACTICAL QUESTION PAPER

BOTP 102: Paper– I: MICROBIAL DIVERSITY, MYCOLOGY AND PHYCOLOGY

Time 3 Hours

Max Mars - 40

1.	Identify & classify the specimens A, B, & C with labelled diagrams and reasons	3 x 3 = 9 Marks
2.	Identify and Comment on the instrument / Disease/Lichen D	1 x 3 = 3 Marks
3.	Prepare a temporary slide of E . Sketch, label and identify with reasons. Leave the preparation for evaluation	1 x 5 = 5 Marks
4.	Stain the given material F by Gram staining. Write the procedure and identify with reasons. Leave the preparation for evaluation OR Calculate the population of fungal spores /yeast cells in F using Haemocytometer	1 x 6 = 6 Marks
5.	Identify the Slides G, H and I with labelled diagrams with reasons.	3 x 3 = 9 Marks
6.	Record and Submission	5 + 3 = 8 Marks

SCHEME OF EVALUATION

1	Three specimens A, B and C , - one from A - Fungi, B - Cyanobacteria C - Algae, (Identification – 1 mark, classification – 1 mark, reasons 1 mark)
2	Instrument/Disease/Lichens of D – 3 marks, (Identification 1 mark, working principle 1 Mark, Labelled diagram 1mark) /Diseases/Lichens(Identification 1 mark, Comment and Labelled diagram 2 marks)
3	Specimen E from algae –(mounting – 2 marks. Identification – 1 mark, Sketch with reasons 2 marks)
4	Specimen F – Gram staining (Staining – 4 marks, Procedure and result –2 mark). OR Calculation of fungal spores/yeast cells using haemocytometer (Procedure -2 marks, observation – 2 marks, calculation – 2 marks)
5	Three permanent slides G,H & I - from Fungi/Cyanobacteria / algae (Identification – 1 mark, sketch with reasons- 2marks)
6	a) Record –5 marks b) Submission of 3 specimens –3 marks

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BOTANY CURRICULUM

II SEMESTER

PAPER – II: BRYOPHYTES, PTERIDOPHYTES, PALEOBOTANY AND PLANT ANATOMY

Programme name	B. Sc. BOTANY	SEMESTER	II
Course Title	PAPER – II: BRYOPHYTES, PTERIDOPHYTES, PALEOBOTANY AND PLANT ANATOMY		
Course Code	BOTT - 201	No. of Credits	03
Contact Hours	60 Hours	Duration of Exam	03 Hours
Formative Assessment Marks	20	Summative Assessment Marks	80

Course objectives

Understand the basics of Bryophytes and Pteridophytes covering the characteristics, occurrence, organization, and classification of Bryophytes and Pteridophytes.

Explore the economic and ecological roles of Bryophytes and Pteridophytes in the field of medicine, agriculture and industry.

Investigate the structure, reproduction and life cycles of Bryophytes and Pteridophytes.

To understand the mode of evolution of plants through paleobotanical studies, formation of fossils and type study.

Course outcomes:

Understand the diversity and affinities among Bryophytes and Pteridophytes.

Understand the morphology, anatomy, reproduction and life cycle across Bryophytes and Pteridophytes and their economic, ecological and evolutionary significance.

Observation of fossil types, their formation and structure of various plant parts in different era.

Skill development for the proper description of internal structure using

II SEMESTER

PAPER – II: BRYOPHYTES, PTERIDOPHYTES, PALEOBOTANY AND PLANT ANATOMY

60 Hrs.

Unit I	BRYOPHYTES General characters, classification, distribution, structure, reproduction and alternation of generations in <i>Marchantia</i> , <i>Anthoceros</i> and <i>Funaria</i> . Ecological and economic importance of Bryophytes.	15 Hrs.
Unit II	PTERIDOPHYTES Introduction and general characters of Pteridophytes Study of diversity in morphology, anatomy, reproduction and life cycle of the following groups in representative forms: 1. Psilotopsida – Ex. <i>Psilotum</i> 2. Lycopsidea – Ex. <i>Selaginella</i> 3. Filicopsida - Ex. <i>Marsilea</i> (Developmental stages not required) Brief account of stelar evolution, heterospory and seed habit.	15 Hrs.
Unit III	PALEOBOTANY Contributions of paleobotanist – Birbal Sahni. Outline of geological time scale with special emphasis on Paleozoic and Mesozoic Era. Process of fossilization – Compression, Impression, Petrification, Compaction, Casts and Moulds, Coal balls. Type study – <i>Rhynia</i> , <i>Cycadeoidea</i> and <i>Pentaxylon</i> .	15 Hrs.
Unit IV	Meristematic Tissues - Structure, function and classification. Organisation of Apical Meristems: Apical cell theory, Tunica-corpus theory and Histogen theory. Shoot and root apical meristems. Histology: Structure, Classification and significance of simple, complex and secretory tissues. Types of vascular bundles. Secondary growth: Dicot stem Ex. – <i>Tridax</i> . Anomalous Secondary growth: <i>Boerhaavia</i> and <i>Dracaena</i> . Brief account of wood anatomy	15 Hrs.

Assessment:	Marks:
Attendance	05 Marks
Assignment	05 Marks
Test	10 Marks
Total	20 Marks

II SEMESTER

PAPER – II: BRYOPHYTES, PTERIDOPHYTES, PALEOBOTANY AND PLANT ANATOMY

No. of credits: 02

45 Hrs.

	BOTP – 202: BRYOPHYTES, PTERIDOPHYTES, PALEOBOTANY AND PLANT ANATOMY	15 Units
1.	Study of Bryophytes (Forms studied in theory)	3 Units
2	Study of Pteridophytes (Forms studied in theory)	4 Units
3	Paleobotany – Type study (Fossil material/slide/photos)	2 Units
4	Histology – Permanent tissues	2 Units
5	Study of T. S of Dicot stem and sectioning: <i>Tridax</i>	1 Unit
6	Study of T. S of <i>Boerhaaviastem</i> and sectioning.	1 Unit
7	Study of T. S of <i>Dracaena</i> stem and sectioning.	1 Unit
8	Visit to Institute of Wood Science	1 Unit

Assessment:	Marks:
Continuous assessment/Attendance	05 Marks
Test	05 Marks
Total	10 Marks

PRACTICAL QUESTION PAPER

BOTP 202: PAPER – II: BRYOPHYTES, PTERIDOPHYTES, PALEOBOTANY AND PLANT ANATOMY

Time: 3Hrs

Marks: 40

1.	Identify and classify the specimens A, B, C and D giving reasons	4 x 3 = 12 Marks
2.	Comment on Slide/ Specimen/ Photograph of E	1 x 3 = 3 Marks
3.	Identify the slides F, G, H and I with reasons and diagrams	4 x 3 = 12 Marks
4.	Prepare a temporary stained T. S of the material J . Sketch, label and identify with reasons. Leave the preparation for evaluation.	1 x 5 = 5 Marks
5.	Record and report of visit to Wood Science Institute	5 + 3 = 8 Marks

SCHEME OF EVALUATION

1	A&B Bryophytes and C&D Pteridophytes: (Identification and Classification- 1 mark, Reasons- 2 marks)
2	Slide/ Specimen/ Photograph of E. (Paleobotany) (Identification – 1 mark, Reasons- 1 mark and Diagram – 1 mark)
3	Slides F (Bryophytes), G (Pteridophytes), H (Paleobotany) and I Anatomy (Identification – 1 mark, Reasons- 1 mark and Diagram – 1 mark) for each slide.
4	J (Plant Anatomy) – (Staining and mounting – 2 marks, Sketch and labelling – 1 mark, Identification – 1 mark and reasons – 1 mark)
5	a) Record – 5 marks. b) visit to Wood Science Institute – 3 marks

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QUESTION PAPER FORMAT THEORY EXAMINATION

Marks for each question	No. of questions to be		Total marks
	Answered	Out of	
A. 2	10	12	20
B. 5	6	8	30
C. 10	3	5	30
			80

B.Sc. Degree Examination December/January 2025

(Undergraduate Credit Based Semester Scheme) BOTANY

BOTT: 101- PAPER-I: Microbial Diversity, Mycology and Phycology

Time: 3 Hours

Max. Marks: 80

Section-A

A. Explain/define **any ten** of the following in **two** or **three** sentences: **(10 X 2 = 20)**

1.

2.

3.

4.

5.

6.

7.

8.

9.

10.

11.

12.

Section- B

B. Write critical notes on **any six** of the following:

(6 X 5 = 30)

13.

14.

15.

16.

17.

18.

19.

20.

Section- C

C. Give a comprehensive account on **any three** of the following:

(3 X 10 = 30)

21.

22.

23.

24.

25.

Question Paper Format for Discipline Specific (Elective) Subjects

THEORY EXAMINATION

Marks for each question	No. of questions to be		Total marks
	Answered	Out of	
A. 2	5	7	10
B. 5	6	8	30
			40

B.Sc Degree Examination December/January 2025

(Undergraduate Credit Based Semester Scheme)

BOTANY

BOTT: Paper: Elective

Time: 2 Hours

Max Marks:

40

Section-A

A. Explain/define **any ten** of the following in **two** or **three** sentences: (5 X 2 = 10) 1.

- 2.
- 3.
- 4.
- 5.
- 6.
- 7.

Section- B

B. Write critical notes on **any six** of the following: (6 X 5 = 30) 8.

- 9.
- 10.
- 11.
- 12.
- 13.
- 14.
- 15.

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DEPARTMENT OF BOTANY



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**Syllabus for III & IV SEMESTERS B.Sc. Botany (UG)
For the academic year 2025-26**

CHOICE BASED CREDIT SYSTEM (CBCS)

Framed According to the State Educational Policy (SEP 2024)

[To be implemented from the academic year 2025-26]

BENGALURU CITY UNIVERSITY

CHOICE BASED CREDIT SYSTEM
(As per SEP)

Syllabus for Botany

2025-26

Board of Studies in Botany (UG),
held on 29. 05. 2025 at 11.00 AM at BCU,

/2025-26 Dated 20.05.2025

AGENDA

(UG) programme (III and IV Semesters) of BCU,

and placed the agenda for discussion and approval.

discussed and prepared the new syllabus for B.Sc.,

programme of Bengaluru City University, Bengaluru

current academic year (2025-26). The board has

as for B.Sc., Botany programme for its introduction

lly, the chairman thanked the members of the board

nosperms And ryology of osperms	4 Hrs.	3 Credits	80+20
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MEMBERS PRESENT

- 1) Prof. G. R. Janardhana (Chairman)**
Dept. of Studies in Botany
University of Mysore, Mysore, Mysore-06
- 2) Prof. L. Rajanna (Member)**
Senior Professor, Dept. of Botany
Bangalore University, Bangalore-56
- 3) Mrs. Shobharani (Member)**
Associate Professor, Govt. First Grade College
Yelahanka, Bengaluru-64
- 4) Mrs. Esther Watson (Member)**
Dept. of Botany, Bishop Cotton College for Women
Bengaluru-27
- 5) Mrs. Roopashree (Member)**
Dept. of Botany, KLEs Nijalingappa College
Bengaluru-10.
- 6) Dr. P.B. Mallikarjuna (Member)**
Professor, Govt. First Grade College

Sem	CourseCode	Practical/ Theory	TITLE OF THE COURSES	Hrs./ Week L:T:P	Credits	Max Marks
I	DSCB*-1.1	Theory	Microbial Diversity, Mycology and Phycology	4 Hrs.	3	80+20
		Practical	Microbial Diversity, Mycology and Phycology	3 Hrs.	2	40+10
II	DSCB*-1.2	Theory	Bryophytes, Pteridophytes, Paleobotany and Plant Anatomy	4 Hrs.	3	80+20
		Practical	Bryophytes, Pteridophytes, Paleobotany and Plant Anatomy	3 Hrs.	2	40+10
III	DSCB*-1.3	Theory	Gymnosperms And Embryology of Angiosperms	4 Hrs.	3	80+20
		Practical	Gymnosperms And Embryology of Angiosperms	3 Hrs.	2	40+10
	DSEB-1.1	Elective	Mushroom Cultivation Technology	2 Hrs.	2	40+10
	DSEB-1.2	Elective	Plant Propagation	2 Hrs.	2	40+10
IV	DSCB*-1.4	Theory	Cytology Genetics and Molecular Biology	4 Hrs.	3	80+20
		Practical	Cytology, Genetics and Molecular Biology	3 Hrs.	2	40+10
	DCB**-1.1	Compulsory Practical	Nursery, Gardening & Landscaping	4 Hrs.	2	40+10
	DSEB-1.1	Elective	Medicinal Plants and Herbal Technology	2 Hrs.	2	40+10
	DSEB-1.2	Elective	Medicinal Plants	2 Hrs.	2	
V	DSCB-1.5	Theory		4 Hrs.	3	80+20
	DCB**-1.2	Practical		3 Hrs.	2	40+10
	DSEB-1.4	Theory		3 Hrs.	2	40+10
VI	DSCB-1.6	Theory		4 Hrs.	3	80+20
	DCB**-1.3	Practical		3 Hrs.	2	40+10
	DSEB-1.5	Theory		3 Hrs.	2	40+10

DSCB* Discipline Core Course Botany; **DSEB*** Discipline Elective Botany;

DCB* Discipline Compulsory Course Botany; EC-Open Elective Course.

III SEMESTER**DSCB-1.3: GYMNOSPERMS AND EMBRYOLOGY OF ANGIOSPERMS**

PROGRAMME	B.Sc., BOTANY	SEMESTER	III
Course Title	DSCB-1.1: GYMNOSPERMS AND EMBRYOLOGY OF ANGIOSPERMS		
Course Code		No. of Credits	03
Contact Hours	56 Hours	Duration of Exam	03 Hours
Formative Assessment	20 Marks	Summative Assessment	80 Marks

Course Objectives: The course intends to train the students to give better understanding the diversity, morphology and distribution and classification of Gymnosperms. Further it also provides the students the life cycles and their economic importance. The course also trains the students on the developmental biology of aspects of angiosperms. And also, the applications of these phenomenon in understanding the all aspects of plant development.

Course Outcome: After the completion of the course, the students are able to explain the diversity, distribution and classification system used study Gymnosperms, morphology, anatomy, reproduction and lifecycles of Gymnosperms. Further, the students will be able to explain the key concepts and mechanisms involved in the reproductive biology of angiosperms, including flower development, pollination, fertilization, and seed formation. Overall, the course will provide the learners with the knowledge required to analyse and apply concepts in plant development, breeding, and biotechnology.

UNIT I	INTRODUCTION TO GYMNOSPERMS	16 Hrs.
	General characteristics, distribution and classification of Gymnosperms. Morphology and anatomy of root, stem (primary and secondary growth) and leaf, Reproductive structures (Developmental Stages not required) and life cycles of <i>Cycas</i> , <i>Pinus</i> and <i>Gnetum</i> . Angiosperm characteristics and evolutionary significance of <i>Gnetum</i> . Phylogeny of Gymnosperms. Economic importance of Gymnosperms.	

UNIT II	EMBRYOLOGY OF ANGIOSPERMS Contribution of Indian botanists to Plant Embryology – P. Maheshwari, B. M. Johri, BGL Swamy. Structure of typical Angiosperm flower. Structure of stamen. Microsporangium- Early stages of development of anther (Primordial stage to differentiation of wall layers and sporogenous tissue), Structure of young and mature anther. Tapetum-Origin, types, structure & functions. Microsporogenesis - Microspore mother cell stage, meiocyte stage, cytokinesis, microspore tetrads. Abnormalities-Compound pollen grains. Pollinia. Microgametogenesis –Formation of vegetative and generative cells, structure of male gametophyte. Abnormalities – Nemec phenomenon.	16 Hrs
UNIT III	EMBRYOLOGY OF ANGIOSPERMS (Contd.) Structure of Pistil - Placentation- definition and types. Megasporangium –Structure of ovule – Integuments (endothelium), Micropyle (Obturator), Nucellus (Crassinucellate and Tenuinucellate conditions). Types of ovules -Anatropous, Orthotropous, Amphitropous, Circinotropous. Megasporogenesis & Megagametogenesis –Development of Female gametophyte. Embryosac- monosporic- <i>Polygonum</i> type, bisporic- <i>Allium</i> type, tetrasporic- <i>Fritillaria</i> type. Structure of mature embryo sac. Double Fertilization – pollen germination, growth of pollen tube through style (solid and hollow styles), entry of pollen tube into ovule (porogamy, mesogamy, chalazogamy), entry of pollen tube into the embryosac, pollen tube discharge, syngamy, triple fusion. Significance of double fertilization, post fertilization changes.	16 Hrs
UNIT IV	PALYNOLOGY AND EXPERIMENTAL EMBRYOLOGY Dicot embryo development. Structure of Dicot and Monocot embryo (grass). Parthenocarpy - definition. Polyembryony and Apomixis and their significance. Seed – Structure of Dicot and Monocot seed. PALYNOLOGY - Pollen morphology –shape, size and apertures, NPC system, pollen wall stratification., pollen Kit. Applied Palynology – Aeropalynology, Mellisso palynology. EXPERIMENTAL EMBRYOLOGY –A brief account on Experimental embryology. Totipotency, Basic steps in plant tissue culture technique. MS Media and its basic components. Tissue culture techniques and their practical applications-Anther culture, Embryo culture, Protoplast culture.	16Hrs

Theory Formative Assessment (Internal Assessment)	Marks
Attendance	05 Marks
Assignment	05 Marks
Test	10 Marks
TOTAL	20 Marks

THEORY QUESTION PAPER PATTERN FOR EXAMINATION

Credits:03

80 Marks

Marks for each question	No. of Questions		Total Marks
	To be answered	Out of	
A.2Very short answers	10	12	20
B. 5 Short notes	6	8	30
C.10 Essay types	3	5	30
TOTAL			80 Marks

III SEMESTER**DSCB-1.3:GYMNOSPERMS AND EMBRYOLOGY OF ANGIOSPERMS****No of Credits:2****PRACTICAL****42 Hrs**

Sl. No.	CONTENTS	15Units
1	Study of materials and permanent slides of Gymnosperms included in theory	6 Units
2	Study of permanent slides of microsporogenesis and male gametophyte	1Units
3	Mounting of Pollen grains— <i>Hibiscus</i> , Grass, <i>Mimosa</i> . Mounting Pollinia of <i>Calotropis</i>	1Units
4	Pollen germination (hanging drop method).	1Units
5	Permanent slides of types of ovules, Megasporogenesis & embryo sac development	1Unit
6	Permanent slides of types of placentation—Axile, Marginal, Parietal, basal types. Sectioning of ovary, for any two types of placentation.	1Unit
7	Mounting of embryo- <i>Tridax</i>	1Unit
8	Mounting of endosperm- <i>Cucumis</i> .	1Unit
9	Mini project work in groups of 3-5 students, from the following list. a) Study of pollen morphology of different flowers with respect to shape, colour, pores etc. b) Pollen germination of different pollen grains and calculation of percentage of germination. c) Calculating percentage of germination of one particular type of pollen grain collected from different localities/ under different conditions. d) Study of placentation of different flowers. e) Any other relevant study related to Gymnosperms/Embryology. Mini project work may be carried out in groups of 3-5 students, supervised by the batch in-charge teacher. The mini project report of 5-6 pages (hand written/ typed), to be submitted in the practical examination.	2Unit

Practical Formative Assessment (Internal Assessment)	Marks
Continuous assessment/Attendance	05 Marks
Practical Test	05 Marks
TOTAL	10 Marks

PRACTICAL QUESTION PAPER PATTERN FOR EXAMINATION**Time-03Hrs****Marks– 40**

1	Identify and classify specimens A, B and C giving reasons.	3X3=09
2	Identify the slides D, E, F and G with reasons and labeled diagrams.	3X4=12
3	Mount the embryo/ Endosperm of specimen G & comment.	1X5=05
4	Mount the pollinia/ perform pollen germination of specimen H & comment.	1X4=04
5	Record & Project submission	5+5=10

Suggested Readings

- 1) Bhatnagar, S. P. and Moitra, A.1971.Gymnosperms,New Age Publications, New Delhi
- 2) Bhojwani,S.S. and Bhatnagar, S. P 1974. The Embryology of Angiosperms, Vikas Pub, New Delhi.
- 3) Chopra, G. L. and Verma, V. 1988.Gymnosperms. Pradeep Publications, Jalandar, India.
- 4) Coulter and Chamberline, 1964. Morphology of Gymnosperms, Central Book Depot. New Delhi.
- 5) Davis, G.L.1966. Systemic Embryology of Angiosperms. John Wiley & Sons Inc, New York.
- 6) Johansen, D.A.1950. Plant Embryogeny. Chronica Botanica Co., Waitham, Mass.
- 7) Johri, B. M 1984. Embryology of Angiosperms. Springer–Verlag, Berlin.
- 8) Maheshwari, P. 1950. An Introduction to Embryology of Angiosperms. Tata Mc Graw Hill, New York.
- 9) Maheshwari, P. 1963. Recent Advances in Embryology, Intl. Soc. Pl. Morphol. NewYork.
- 10) Pullaiah, T., Lakshminarayana, K., & Rao, B. H. (2019). *Plant Reproduction 2nd Ed.* Scientific Publishers.

- 11) Raghavan, V. 1979. Experimental Embryogenesis of Vascular Plants, Cambridge University Press, Cambridge, U.K.
- 12) Sporne, K. R 1974. The Morphology of Gymnosperms. Hutchins on & Co., London
- 13) Swamy B. G. L. and Krishnamurthy K. V, 1980. From flower to fruit: embryology of flowering plants. Tata Mc Grew- Hill Pub Co. New York.
- 14) Vashista, B. R. 1990. Gymnosperms. S C hand & Co., Pvt. Ltd., New Delhi.
- 15) Vasishta, D. P., Sinha, D. A., & Kumar, D. A. (2017). Botany for degree students Gymnosperms. S.Chand Publishing.

III SEMESTER**DSEB-1.1: MUSHROOM CULTIVATION TECHNOLOGY**

PROGRAMME	B.Sc.,BOTANY	SEMESTER	III
Course Title	DSEB: MUSHROOM CULTIVATION TECHNOLOGY		
Course Code		No. of Credits	02
Contact Hours	28	Duration of Exam	1.5 Hours
Formative Assessment	10	Summative Assessment	40 Marks

Learning Objectives: The course intends to train students on edible mushrooms. The course teaches students on science of edible mushrooms, types, life cycles and economic importance of mushrooms. Course will also teach health and economic benefits of mushrooms. The course will also help them understand the medicinal and nutritive value of mushrooms.

Learning Outcomes: After successful completion of the course, the students will be able to practice the techniques of cultivation of edible mushrooms. They will be able to setup entrepreneurial small-scale units for mushroom cultivation for self-employment. Students will be able to use the practical skills for large-scale industrial multiplication of edible mushrooms.

UNIT-I	Mushrooms Biology Introduction to Fungi. Basidiomycetes -forming mushrooms. Distinguishing characteristics of mushrooms and their Life cycles. Variations in Morphology and classification of mushrooms. History of mushroom cultivation. Mushroom industry in India, Food values of Mushrooms, Uses of mushrooms. Edible and poisonous mushrooms. Medicinal mushrooms. Economic importance of mushrooms.	16Hrs.
UNIT-II	Mushroom Cultivation Methods Types of edible mushrooms. Button, Straw and Oyster mushroom – General morphology. Substrates for cultivation. Composting, bed preparation, sterilization of substrates. Spawn production: Preparation of spawn substrate, obtaining pure culture. Media used in raising pure culture. Maintenance of mother culture. Inoculation and Incubation.	16Hrs

UNIT-III	Cultivation of Button mushroom (<i>Agaricus bisporous</i>), Oyster mushroom (<i>Pleurotus sajor-caju</i>), Paddy straw mushroom (<i>Volvariella volvaceae</i>). Precautions of mushroom cultivation. Post- Harvest Technology of Mushrooms Harvesting and Processing of mushrooms. Storage of mushrooms. Marketing of Mushrooms. Diseases and pest management during mushroom cultivation. Recipes of Mushroom. Value added mushroom products. A brief account on cultivation of medicinal mushrooms(Cordyceps).	16 Hrs
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Theory Formative Assessment (Internal Assessment)	Marks
Attendance	05 Marks
Visit to mushroom cultivation unit	05 Marks
TOTAL	10 Marks

THEORY QUESTION PAPER PATTERN FOR EXAMINATION

Time -1.5 Hrs

40 Marks

Marks for each question	No. of questions to be		Total Marks
	Answered	Out of	
PART-A(2 Marks)	5	7	10
PART -B (5 Marks)	6	8	30
TOTAL			40

Suggested Readings

- 1) Nita, B. (2000). Handbook of Mushrooms. Oxford and IBH Publishing Co. Pvt. Ltd., New Delhi.
- 2) Pandey, R. K. and Ghosh, S. K.(1996). A hand book of Mushroom Cultivation. Emkey Publication.
- 3) Pathak,V. N. and Yadav, N. (1998). Mushroom Production and Processing Technology. Agrobios, Jodhpur.
- 4) Tewari, P. and Kapoor S. C. (1998). Mushroom Cultivation, Mittal Publication, New Delhi.
- 5) Dubey, R. C., 2005 A Text book of Biotechnology S. Chand & Co, New Delhi.

- 1) Dey, Kalyan Kumar (1992). An Introduction to Plant Tissue Culture, New Central Book Agency, Das Lane, Calcutta.
- 2) M. K. Sadhu. 2017 (Reprint) Fundamentals of Horticulture and Plant Propagation.
Emphasizes propagation under Indian conditions.
- 3) De.P.C.2015 (Revised Edition) Plant Propagation and Nursery Management. Useful for understanding both propagation techniques and nursery practices.

IV SEMESTER

DSCB-1.4: CYTOLOGY, GENETICS AND MOLECULAR BIOLOGY

PROGRAMME	B.Sc.,BOTANY	SEMESTER	IV
Course Title	DSCB-1.4:CYTOLOGY GENETICS AND MOLECULAR BIOLOGY		
Course Code		No.of Credits	03
Contact Hours	56 Hours	Duration of Exam	03 Hours
Formative Assessment	20 Marks	Summative Assessment	80 Marks

Learning Objectives: The course would help the students understand the principles of genetics, the mechanisms of heredity, variation, and the molecular basis of gene function. The course fosters skills in interpreting genetic data analysis. The molecular biology course aims to provide students with a comprehensive understanding of structure and functions of nucleic acids. Gene expressions, cellular structures and the molecular mechanisms that govern key processes such as DNA replication, transcription, translation, and cell signaling. Overall, it lays a strong foundation for further studies or Careers in biomedical research, biotechnology, and health sciences.

Course Outcome: Upon completing the course, the students will be able to explain the structure and function of cellular organelles and understand the molecular basis of key cellular processes such as structure and functions of DNA and RNA, gene expression, cell signalling, and cell division etc. They will be able to demonstrate the ability to apply fundamental laboratory techniques and analyze data from molecular and cellular experiments.

UNIT I	Plant cell – Ultrastructure, cell organelles – Structure and function (Nucleus, Endoplasmic reticulum, Golgi complex, Vacuole, Mitochondria and Chloroplast), Cytoskeleton. Structure of Eukaryotic Chromosome, - centromere, Kinetochore and telomere. Structural organization - nucleosome model. Types of Chromosomes – based on centromere position and number. B-chromosomes and their significance. Cell cycle– Phases of eukaryotic cell cycle. Cell division–Mitosis and meiosis and Its significance. Karyotype– Types and significance. Programmed Cell Death(PCD).	16Hrs
UNIT II	Mendelian Genetics – Introduction, history, Terminologies – Factors /Alleles, phenotype, genotype, homozygous, heterozygous, true breeding, test cross, back cross. Mendelian pattern of inheritance- Monohybrid and dihybrid crosses. Laws based on monohybrid and dihybrid cross. Non-Mendelian genetics – Incomplete Dominance. Interaction of genes - complementary, supplementary factors/ recessive epistasis, dominant epistasis. Polygenic inheritance or multiple factor inheritance. Genetic problems.	16Hrs

UNIT III	Linkage and its types. Crossing over – meaning, mechanism and types. Sex Determination in <i>Melandrium</i> . Variation in chromosome number (Euploidy. Polyploids – Autopolyploids and Allopolyploids. Aneuploidy – Hypoploidy and hyperploidy). Variation in chromosome structure (deletions / deficiency, duplications, inversions and translocation). Gene mutations–Types, Molecular basis of Mutations; Mutagens–physical and chemical (Base analogs, deaminating, Alkylating and intercalating agents). Fine structure of gene	16Hrs
UNIT IV	Molecular Biology – Historical perspectives, discovery of DNA as the genetic material (Griffith's, Harshey and Chase experiments). Nucleic acids – DNA structure, composition, types and the mechanism of replication. A brief account of DNA repair mechanism. RNA – Structure, composition and types. Central dogma of Molecular biology ,genetic code–Salient features. Protein synthesis (Transcription and translation).Gene regulation in prokaryotes(Lac Operon).	16 Hrs

Practical Formative Assessment (Internal Assessment)	Marks
Continuous assessment/Attendance	05 Marks
Test	05 Marks
TOTAL	10 Marks

THEORY QUESTION PAPER PATTERN FOR EXAMINATION

Marks for each question	No. of Questions		Total Marks
	To be answered	Out of	
A.2Veryshort answers	10	12	20
B. 5 Short notes	6	8	30
C.10 Essay types	3	5	30
TOTAL			80 Marks

SEMESTER IV**DSCB:CYTOLOGY GENETICS AND MOLECULAR BIOLOGY****PRACTICALS****Credits:2****42 Hrs**

Sl. No.	CONTENTS	15Units
1	Study of cell and its organelles with help of electron micrographs	1 Unit
2	Preparation of cytological stains – Acetocarmine and Acetoorcein	1 Unit
3	Study of permanent slides of Mitosis and Meiosis	2 Units
4	Mitotic squash preparation and study of stages from – Onion roots	2 Units
5	Meiotic stage preparation and study of stages from –Onion flower buds	2 Units
6	Karyo typing in <i>Alumcepa</i> .	1 Unit
7	Solving Genetic problems related Mendelian and Non-Mendelian Genetics	4 Units
8	Cytology- permanent slides preparation and observations	2 Units
9	Isolation of genomic DNA from plant samples	Unit 4
10	Isolation of plasmids from <i>E. coli</i> .	Unit 4
11	Estimation of DNA	Unit 4
12	Estimation of RNA	Unit 4
13	Demonstration of Restriction digestion	Unit 4
14	Demonstration of DNA ligation	Unit 4
15	Separation of nucleic acids by Agarose gel electrophoresis.	Unit 4
16	Study of Photographs, charts of Cytologists and Molecular Biologists	Unit 4

PRACTICAL QUESTION PAPER PATTERN FOR EXAMINATION**Time-03Hrs****Marks-40**

1	Prepare of Mitotic squash of material 'A'. Identify the stage, sketch and give reason.	1X6=6
2	Prepare of Meiotic smear of material 'B'. Identify the stage sketch and give reason.	1X6=6
3	Identify the slides C and D, sketch and comment	2X3=6
4	Identify and comment on E and F	2X3=6
5	Solve the genetic problem/s G	1X6=6
6	Record & Permanent Slides Submission	5+5=10

Suggested Readings

- 1) Cooper, G.M., Hausman, R.E. (2009). The Cell: A Molecular Approach, 5th edition. Washington DC: ASM Press & Sunderland, Sinauer Associates, MA.
- 2) Karp. G. (2010). Cell Biology, 6th edition. New Jersey, U. S. A. John Wiley & Sons.

- 3) De Robertis, E. D. P. and De Robertis R.E. 2009. Cell and Molecular Biology, 8th edition.
- 4) Lippincott Williams and Wilkins, Philadelphia.
- 5) Becker W. M., Klein smith L J, and Bertni G. P .2009. The World of the Cell. 7th edition. Pearson, Benjamin Cummings Publishing, San Francisco.
- 6) Reven, F.H., Evert, R.F., Eichhorn, S.E. (1992). Biology of Plants. New York, NY: W.H. Freeman and Company.
- 7) Alberts, B., Bray, D., Hopkin, K., Johnson, A. D., Lewis, J., Raff, M., Roberts, K., & Walter, P. (2013). Essential cell biology (4th ed.). Garland Publishing.
- 8) Verma, P. S. (2004). Cell Biology, Genetics, Molecular Biology: Evoloution and Ecology, India: S. C hand Limited.
- 9) Gardner, E.J., Simmons, M. J., Snustad, D. P. (1991). Principles of Genetics, 8th edition. New Delhi, Delhi, John Wiley & sons.
- 10) Griffiths, A.J.F., Wessler, S.R., Carroll, S.B., Doebley, J. (2010). Introduction to Genetic Analysis, 10th edn. New York, NY: W.H. Freeman and Co.

IV SEMESTER
CPC-1.1:NURSERY, GARDENING AND LAND SCAPING
(Compulsory Practical)

PROGRAMME	B.Sc.,BOTANY	SEMESTER	IV
Course Title	CPC-1.1:NURSERY GARDENING AND LANDSCAPING		
Course Code		Number of Credits	2
Contact Hours	42	Duration of Exam	3Hrs
Formative Assessment Mark	10	Summative Assessment Marks	40

Learning Objectives: The course is aimed to teach students the basic knowledge required to develop entrepreneurship skills for the development of Nursery, Gardening and Landscaping. This course would train students to initiate remunerative enterprise owing to a high demand of skilled Professionals in this field.

Course Outcome: After the completion of the course, the learners will be able to understand the importance of nursery in gardening. Students will also be able to describe and differentiate between the types of gardens in the city areas. They will be able to execute several nursery and gardening operations in clouding media and equipment's needed for the establishment of gardens. They will also be a position to conduct and learn different methods for plant adopted from different plants. Learners will also assess the growing conditions of different horticultural plants, their general requirements and understand their role in landscaping.

COMPULSORY PRACTICALS

1. Media for propagation of plants in Nursery Beds, Pots and Mist chamber.
2. Methods of preparation of nursery beds and sowing of seeds.
3. Study of different propagation methods viz., cutting, layering, division, grafting and budding in different ornamental plants.
4. Introduction and practicing Bonsai training, pruning and wiring.
5. Study of different types of gardens (indoor and outdoor) and key features of gardens (Paths & Avenues, Hedges & Edges, Lawn, Flowerbeds, Arches & Pergolas, Fencing, Water bodies, Rock Garden).
6. Study of Vertical Gardening and Roof top gardening
7. Methods for selection and enlisting of suitable plants for different locations and in different types of gardens.

8. Identification of key horticultural plants, Herbs including different types of grasses – foliage and flowering, Shrubs including hedge plants – foliage and flowering, Avenue trees – foliage and flowering, Climbers, Lianas, Epiphytes, Creepers, Trailers, Aquatic plants, Succulents, Weeds.
9. Study of important gardens of India (any five).
10. Methods of Landscape designing of Residential areas and Public Gardens, Aquatic Garden, Rock Garden, Industrial gardens.
11. Concept and Application of Computer aided Designing (CAD) for landscape designing/
12. Preparation of landscape designs for school and college using CAD technology.
13. Study of Interior designing with plants - Terrarium, miniature landscape
14. Preparation and application of herbal pesticides
15. Demonstration of different composting methods.

Theory Formative Assessment (Internal Assessment)	Marks
Attendance	05 Marks
Assignment	05 Marks
TOTAL	10 Marks

SCHEME OF PRACTICAL QUESTION PAPER

Time-03 Hrs

40 Marks

Q.No.	Question Paper Pattern	Marks
1	Perform the experiment A	10 Marks
2	Perform the experiment B and C (2x4)	08 Marks
3	Comment on C, D and E (4x3)	12 Marks
4	Prepare a temporary Terrarium/Miniature Landscape F	05 Marks
5	Class Record	05 Marks

Suggested Readings:

- 1) Ah and book of Landscape: CPWD
- 2) Gopalaswamy Iyengar, K. S., Parthasarathy, G., Mukundan, P. (1991). Complete Gardening in India. India: Gopalaswamy Parthasarathy, 'Srinivasa'.
- 3) Hartmann, H. T., Kester, D.E., Hartmann, H.T., Kester, D. E. (1975). Plant Propagation: Principles and Practices. India: Prentice-Hall.
- 4) Hodge, G., Hodge, G. (2014). Practical Botany for Gardeners: Over 3,000 Botanical Terms Explained and Explored. United Kingdom: University of Chicago Press
- 5) Littlepage, R., Littlepage, R. (2017). Fundamentals of Garden Design: An Introduction to Landscape Design. (n.p.): Create Space Independent Publishing Platform.
- 6) Roy, R. K., Roy, R. K. (2013). Fundamentals of Garden Designing: A Colour Encyclopedia. India: New India Publishing Agency.
- 7) The Royal Horticultural Society Gardening Manual. (2000). United Kingdom: Dorling Kindersley.

IV SEMESTER DSEB:1.2: MEDICINAL PLANTS

PROGRAMME	B.Sc., BOTANY	SEMESTER	IV
Course Title	DSEB-1.2: MEDICINAL PLANTS		
Course Code		Number of Credits	02
Contact Hours	28	Duration of Exam	1.5 Hrs
Formative Assessment Mark	10	Summative Assessment Marks	40

Learning objectives: The course intends to teach learners about medicinal plants, their identification and classification along with their therapeutic values. It enables them to study the chemical compounds responsible for medicinal effects and their modes of action. Students gain knowledge of traditional and modern uses of medicinal plants in healthcare. This foundation supports research and sustainable use of plant-based medicines for health problems confronting the society.

Learning Outcome: Students are able to understand the medicinal plants world and their importance in day-to-day life. They will be able to understand the ethno botanical value of medicinal use in India for the treatment of human and animal diseases. Able to learn the methods used to extract the active principles for drug development and their validation. Able to understand the role of secondary metabolites and their exploitation, through characterization. They realize the importance of medicinal plants in drug discovery and their application in modern medicine.

UNIT-I	Medicinal Plants & Importance	16 Hrs
	Definition, concept and scope. History and scope of Medicinal Botany. Ethno- medicinal practices; Local (Village, tribal) and traditional (Ayurveda, Siddha, Homeopathy and Unani) systems. Study, survey, documentation and conservation of medicinal plants of Indian sub- continent. Pharmacognosy: Types of formulations (Lehya, Churna, Thaila, Kashayam,) and their analysis for efficacy, shelf life and adulterations. Quality Control: Ensuring the quality and safety of herbal products through various quality control measures.	

UNIT-II	Medicinal Chemistry: Types of secondary metabolites of medicinal importance (Alkaloids, Glycosides, Phenols, Sterols, Terpenes – its extraction (crude and pure form) and its preliminary screening. A brief account of methods of separation of compounds (Chromatographic and Spectrophotometric) and the techniques involved in elucidation of structure of therapeutic compounds.	16 Hrs
UNIT-III	Pharmacology and Drug Discovery: Invitro and invivotoxicity and efficacy studies; Animal models, cell cultures, enzyme inhibition assays and toxicological evaluation - Mutagenicity, teratogenicity, and carcinogenicity. Reverse pharmacology. Recent trends in Medicinal Botany: Drug Discovery pathways.	16 Hrs

Theory Formative Assessment (Internal Assessment)	Marks
Attendance	05 Marks
Assignment	05 Marks
TOTAL	10 Marks

THEORY QUESTION PAPER PATTERN FOR FINAL EXAMINATION

Time-1.5 Hrs

40 Marks

Marks for each question	No. of questions to be		Total Marks
	Answered	Out of	
PART-A (2Marks)	5	7	10
PART -B (5 Marks)	6	8	30
TOTAL			40 Marks

Suggested Readings

- 1) J B Harborne: (1984): Phytochemical methods; A guide to modern techniques of Plant Analysis. Chapman and Hall Publishers. 2nd Edition.
- 2) Dey PM and Harborne J B, (1987) Methods in Plant Biochemistry: Academic Press; London,
- 3) Yash P Kalra: (1998) Hand Book of Reference Methods for Plant Analysis; CRC Press; London.
- 4) V. V. Sivarajan & I. Balachandran (1994) Ayurvedic Drugs and their Plant Sources by V. V. Sivarajan & I. Balachandran, Oxford & IBH.
- 5) H. Panda (2002). Medicinal Plants Cultivation & Their Uses. Asia Pacific Business Press.

Sd/-

Prof. G. R. JANARDHANA CHAIRMAN, BOS IN BOTANY, BCU

BOTANY Curriculum

Program Name	B.Sc. in BOTANY	Semester	V
Course Title	Plant Morphology and Taxonomy(Theory)		
Course Code:	DSC – BOT-C9 – T	No. of Credits	04
Contact hours	56 Hours	Duration of SEA/Exam	2 ½ hours
Formative Assessment Marks	40	Summative Assessment Marks	60

Course Pre-requisite(s):

Course Outcomes (COs): After the successful completion of the course, the student will be able to:.

CO1. Understanding the main features in Angiosperm evolution

CO2. Ability to identify, classify and describe a plant in scientific terms, thereby, Identification of plants using dichotomous keys. Skill development in identification and classification of flowering plants.

CO3. Interpret the rules of ICN in botanical nomenclature.

CO4. Classify Plant Systematic and recognize the importance of herbarium and Virtual Herbarium, Evaluate the Important herbaria and botanical gardens

CO5. Recognition of locally available angiosperm families and plants and economically important plants. Appreciation of human activities in conservation of useful plants from the past to the present.

Contents	56 Hrs
Unit 1:	14 hrs
Morphology of Root, Stem and Leaf. Their modifications for various functions. Inflorescence – types, Fruits–types, Structure of flower - Floral diagram and floral formula. Introduction to Taxonomy : History, objectives, scope and relevance of Taxonomy.	

Systems of classification: Artificial, Natural and Phylogenetic ; brief account of Linnaeus', Bentham & Hooker's, Engler and Prantl's system and APG IV System (2016).- Merits and demerits of classifications. Taxonomic Literature : Floras, Monographs and Journals. Herbaria and Botanical Gardens : Important herbaria and Botanical Gardens of the world (Royal Botanic Gardens, Kew, England) and India (Indian Botanical Garden, Calcutta) and their roles. Technique of Herbarium Preparation. Virtual herbarium : E-flora, Documentation.	
Unit 2:	14 hrs
Plant identification: Taxonomic dichotomous keys; intended (yolked) and bracketed keys. (brief account only). Plant descriptions: Common Terminologies used for description of vegetative and reproductive parts of the following families. Study of the diagnostic features of Angiosperm families: Magnoliaceae ,Brassicaceae, Rutaceae , Fabaceae (Sub families : Caesalpinoideae, Mimosoideae, and Papilionoideae), Cucurbitaceae, Apiaceae, Rubiaceae, Asteraceae, Lamiaceae, Euphorbiaceae, Orchidaceae , Cannaceae and Poaceae. Plant Taxonomic Evidences : from palynology, embryology, cytology, phytochemistry and molecular data.	
Unit 3:	14 hrs
Taxonomic Hierarchy: Concept of taxa (family, genus, species); Categories and taxonomic hierarchy; Species concepts (biological, morphological, evolutionary). Rank less system of phylogenetic systematics Botanical Nomenclature : Principles and rules (ICN); Latest code –brief account, Brief account of Ranks of taxa, Type concept (Typification), Rule of priority, Effective and Valid publication, rejection of names, Author citation. Nomenclature of hybrids / cultivated species.	
Unit 4:	14 hrs
Biometrics, Numerical Taxonomy; Phenetics and Cladistics: Characters; Variations; OTUs, character weighting and coding; Cluster analysis; Phenograms, cladograms (definitions and differences). Phylogenetic Systematics: Terms and concepts (primitive and advanced, homology and analogy, parallelism and convergence, monophyly, paraphyly, polyphyly, clades, synapomorphy, symplesiomorphy, apomorphy, lineage sorting, serial homology etc). Origin and evolution of angiosperms: Co-evolution of angiosperms and animals; Methods of illustrating evolutionary relationship (phylogenetic tree, cladogram). Molecular taxonomy: DNA sequence of chloroplast gene (rbcL)	

Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (POs 1-15)

Course Outcomes (COs) / Program Outcomes (POs)	Program Outcomes (POs)														
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Pedagogy: Teaching and learning, Seminar, Assignments, etc

Formative Assessment for Theory	
Assessment Occasion/ type	Marks
Attendance	10
Test	10
Assignments	10
Seminar	10
Total	40 Marks
<i>Formative Assessment as per NEP guidelines are compulsory</i>	

Course Title	Plant Morphology and Taxonomy (Practical)			Practical Credits	02
Course Code	DSC-A10			Contact Hours	52 Hours
Formative Assessment		25 Marks	Summative Assessment		25 Marks
Practical Content					
1. Study of root, stem and leaf structure and modifications. Study of inflorescence types. Study of flower and its parts, Study of fruits. Floral diagram and floral formula. 8 hrs					

Study of families mentioned in theory **preferably two** examples for each family and make suitable diagrams, describe them in technical terms (Description, V.S. of flower, section of ovary, floral diagram/s, floral formula/e and systematic position according to Bentham & Hooker's system of classification) and identify up to species using the flora. **28 hrs**

Identify plants/plant products of economic importance (resource botany) belonging to the families mentioned in the syllabus; with binomial, family and morphology of useful parts.

Red gram, Green gram, Horse gram, Black gram, Bengal gram, Indigo, Tamarind, Bitter gourd, Luffa, Asfoetida, Cumin, Coriander, Coffee, Rubber, Leucas aspera, Tapioca, Ricinus, Rice, Wheat, Ragi, Sugarcane, Annona muricata, Mustard and Ruta graveolens. 16

hrs

Field visit: Local or outside area/ Botanical garden/ tribal settlements

Pedagogy: Teaching and learning, conducting experiments, field visits,

Formative Assessment for Practical	
Assessment Occasion/ type	Marks
Attendance	05
Test	10
Field visit (3 marks), Tour report(2 marks)	05
Submission (Economic Botany)	05
Total	25 Marks
Formative Assessment as per NEP guidelines are compulsory	

References	
1	Baker. H.G. 1970. Plant and Civilization, Wadsworth Publishing Company.
2	Datta S C, <i>Systematic Botany</i> , 4th Ed, Wiley Estern Ltd., New Delhi, 1988.
3	Eames A. J. - <i>Morphology of Angiosperms</i> - Mc Graw Hill, New York.
4	Hall, B.G. (2011). <i>Phylogenetic Trees Made Easy: A How-To Manual</i> . Sinauer Associates, Inc. USA
5	Heywood - <i>Plant taxonomy</i> - Edward Arnold London.
6	Jeffrey C. J. and A. Churchil - <i>An introduction to taxonomy</i> – London.
7	Jeffrey, C. (1982). An Introduction to <i>Plant Taxonomy</i> . Cambridge University Press, Cambridge
8	Judd, W.S., Campbell, C.S., Kellogg, E.A., Stevens, P.F., Donogue, M.J., 2002. <i>Plant Systematics: A Phylogenetic approach</i> , 2nd edition. Sinauer Associates, Inc., USA.
9	Lawrence - <i>Taxonomy of Vascular Plants</i> - Oxford & I B H, New Delhi.
10	Manilal, K.S. and M.S. Muktesh Kumar 1998. <i>A Handbook on Taxonomy Training</i> . DST, New Delhi.
11	Manilal, K.S. and A.K. Pandey, 1996. <i>Taxonomy and Plant Conservation</i> . C.B.S. Publishers & Distributors, New Delhi.
12	Manilal, K.S. 2003. <i>Van Rheede's Hortus Malabaricus. English Edition</i> , with Annotations and Modern Botanical Nomenclature. (12 Vols.) University of Kerala, Trivandrum.
13	Naik V.N., <i>Taxonomy of Angiosperms</i> , 1991. Tata Mcgraw-Hill Pub. Co. Ltd., New Delhi.
14	Pandey, S. N, and S.P. Misra (2008)- <i>Taxonomy of Angiosperms</i> - Ane Books India, New Delhi.
15	Radford A B, W C Dickison, J M Massey & C R Bell, <i>Vascular Plant Systematics</i> , 1974, Harper & Row Publishers, New York.
16	Singh G.2012. <i>Plant systematics: Theory and Practice</i> . Oxford and IBH, Pvt. Ltd., New Delhi.
17	Singh V. & Jain - <i>Taxonomy of Angiosperms</i> - Rastogi Publications, Meerut.
18	Sivarajan V. V - <i>Introduction to Principles of taxonomy</i> - Oxford & I B H New Delhi.
19	Any local/state/regional flora published by BSI or any other agency.

THE NATIONAL COLLEGE AUTONOMOUS BASAVANAGUDI, BENGALURU=

GENERAL PATTERN OF THEORY QUESTION PAPER
(60 marks for semester and Examination with 2 ½ hrs duration)

Part- A

1. Question number 1-06 carries 2 marks each. Answer any 05 questions: 10 marks

Part-B

2. Question number 07-11 carries 05 marks each. Answer any 04 questions: 20 marks

Part- C

3. Question number 12-15 carries 10 marks each. Answer any 03 questions: 30 marks

(Minimum 1 question from each unit and 10 marks question may have sub- questions for 7+3 or 6+6 or 5+5 if necessary)

Total : 60 marks

Note: Proportionate weight-age shall be given to each unit based on number of hours prescribed.

SCHEME OF PRACTICAL EXAMINATION

(Distribution of marks): 25 marks for the Semester end examination 4 hrs

- | | |
|--|---------|
| 1. Identify, classify and describe the specimen A and B taxonomically | 6 Marks |
| 2. Identify the given specimen C with technical description | 4 Marks |
| 3. Write the floral diagram and floral formula of the given specimen D | 2 Marks |
| 4. Identification of specimens E, F and G | 6 Marks |
| 5. Submission (Herbarium) | 2 Marks |
| 6. Submission (Record / Journal) | 5 Marks |

General Instructions :

Q1. Give specimen from Dicotyledons (A) and Monocotyledons (B) Q2.

Give specimen from family they studied (C)

Q3. Give specimen from family they studied (D)

Q4. **Specimen/ Materials** from Root / Stem /Leaf (E) , **Inflorescence / Fruit** (F) and Economic importance (G)

Q5. **Submission of 4 Herbarium** Q6.

Submission of Record/ Journal

Note: Same scheme may be used for IA (Formative Assesment) examination

THE NATIONAL COLLEGE AUTONOMOUS BASAVANAGUDI,BENGALURU=

Genetics and Plant Breeding (Theory)

Program Name	B.Sc. in BOTANY	Semester	V
Course Title	Genetics and Plant Breeding (Theory)		
Course Code:	DSC – BOT-C11 - T	No. of Credits	04
Contact hours	56 Hours	Duration of SEA/Exam	2 .5 hours
Formative Assessment Marks	40	Summative Assessment Marks	60

Course Pre-requisite (s):

Course Outcomes (COs): After the successful completion of the course, the student will be able to:.

CO1. Understand the basics of genetics and plant breeding

CO2. Ability to identify, calculate and describe crossing over, allelic generations and frequencies of recombination.

CO3. Interpret the results of mating and pollinations.

CO4. Classify Plant pollination methods

CO5. Recognition. of modes of inheritance of traits/ phenotypes and Phenotype-genotype correlation.

Contents	56 Hrs
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Unit 1:	14hrs
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Mendelian genetics – History, Laws and concepts

Non-Mendelian genetics- Allelic (Incomplete Dominance and Codominance) and non-allelic gene interaction (complementary, supplementary factors, dominant and recessive epistasis) Multiple alleles.

Extrachromosomal Inheritance- Chloroplast Variegation in Four o'clock plant; Mitochondrial mutations in yeast.

Unit 2:	14hrs
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Linkage, crossing over and chromosome mapping.

Linkage and crossing over- Cytological basis of crossing over;

Recombination frequency, two factor and three factor crosses; Interference and coincidence;

Sex Determination in plants (Melandrium)

Variation in chromosome number and structure:

Gene mutations - Types, Molecular basis of Mutations; Mutagens – physical and chemical (Base analogs, deaminating, alkylating and intercalating agents); Detection of mutations.

Fine structure of gene

Population Genetics- (Allele frequencies, Genotype frequencies, Hardy-Weinberg Law, role of natural selection, mutation,)

Evolutionary Genetics- genetic drift. Genetic variation and Speciation.

Unit 3:	14 hrs
Plant Breeding: Introduction and objectives. Breeding systems: Modes of reproduction in crop plants. Important achievements and undesirable consequences of plant breeding. Methods of crop improvement: Plant Introduction, plant genetic resources; Acclimatization; Selection methods: For self-pollinating and cross pollinating crops Types of Vegetative Propagation in plants; Hybridization: Procedure -advantages and limitations.	
Unit-4	14 hrs
Quantitative inheritance: Concept, mechanism, example of inheritance of Kernel colour in wheat, Monogenic vs polygenic Inheritance. Inbreeding depression and heterosis -History, genetic basis of inbreeding depression and heterosis; Applications. Crop improvement and breeding : Role of mutations; Polyploidy; Distant hybridization and role of biotechnology in crop improvement.	

Pedagogy: Teaching and learning, Seminar, Assignments, etc

Formative Assessment for Theory	
Assessment Occasion/type	Marks
Attendance	10
Test (Objective type)	10
Assignments	10
Seminar	10
Total	40 Marks
Formative Assessment as per guidelines are compulsory	

Program Name	B.Sc. in BOTANY	Semester	V
Course Title	Genetics and Plant Breeding (Practical)	Practical Credits	02
Course Code	DSC – BOT – C12 - P	Contact Hours	4 Hours per week
Formative Assessment	25 Marks	Summative Assessment	25 Marks
Practical Content			

Practical: Plant breeding:

Reproductive biology of self and cross pollinated plants.
 Vegetative reproduction -budding, cutting, grafting and layering .
 Hybridization: Emasculation, bagging, pollination and production of hybrids
 Pollen fertility -Tetrazolium test

Practical: Genetics

Mendel's laws through seed ratios. (Monohybrid and dihybrid cross)
 Incomplete dominance and gene interaction through seed ratios (9:7, 9:6:1, 13:3) 7.. Incomplete dominance and gene interaction through seed ratios (15:1, 12:3:1, 9:3:4)

Pedagogy: Teaching and learning, conducting experiments, field / Lab. visits

Formative Assessment for Practical	
Assessment Occasion/type	Marks
Attendance	05
Test	10
Submission of solved problems	10
Total	25 Marks
<i>Formative Assessment as per guidelines are compulsory</i>	

GENERAL PATTERN OF THEORY QUESTION PAPER

(60 marks for semester end Examination with 2 hrs duration) **Part-A**

1. Question number 1-06 carries 2 marks each. 10 marks

Answer any 05 questions:

Part-B

20 marks

2. Question number 07- 11 carries 05 Marks each.

Answer any 04 questions:

Part-C

3. Question number 12-15 carries 10 Marks each. Answer any 03 questions : 30 marks

(Minimum 1 question from each unit and 10 marks question may have sub-questions for 7+3 or 6+4 or 5+5 if necessary)

Total: 60 Marks **Note: Proportionate weight-age shall be given to each unit based on number of hours prescribed.**

SCHEME OF PRACTICAL EXAMINATION

(distribution of marks): 25 marks for the Semester end examination

- | | |
|---|---------|
| 1. Perform the emasculation / pollen germination of the given sample A | 5 Marks |
| 2. Solve the genetic problem B | 4 marks |
| 3 Identification of Specimen// Photographs of C, D and E | 6 marks |
| 4. Viva Voce | 5 Marks |
| 5. Submission (Record) | 5 Marks |

Total 25

General instructions:

- | | |
|--|---------|
| Q1 Material Cassia// Hibiscus/ etc (A) | 5marks |
| Q2. Genetic problemB | 4marks |
| Q3. Down's, Klinefelter's and Turner's syndromes- any one for C, Translocation Ring, Laggards and Inversion Bridge any one for D and vegetative propagation for E. | 6 marks |
| Q5. Viva voce | 5marks |
| Q6. Submission (Record) | 5marks |

Note: Same Scheme may be used for IA (Formative Assessment) examination

References	
1	Acquaah, G. (2007). Principles of Plant Genetics & Breeding. New Jersey, U.S.: Blackwell Publishing.
2	Singh, B.D. (2005). Plant Breeding: Principles and Methods, 7th edition. New Delhi, Delhi: Kalyani Publishers.
3	Chaudhari, H.K. (1984). Elementary Principles of Plant Breeding, 2nd edition. New Delhi, Delhi: Oxford – IBH.
4	Gardner, E.J., Simmons, M.J., Snustad, D.P. (1991). Principles of Genetics, 8th edition. New Delhi, Delhi: John Wiley & sons
5	Griffiths, A.J.F., Wessler, S.R., Carroll, S.B., Doebley, J. (2010). Introduction to Genetic Analysis, 10th edition. New York, NY: W.H. Freeman and Co.
6	Klug, W.S., Cummings, M.R., Spencer, C.A. (2012). Concepts of Genetics, 10th edition. San Francisco, California: Benjamin Cummings
7	Raven, F.H., Evert, R. F., Eichhorn, S.E. (1992). Biology of Plants. New York, NY: W.H. Freeman and Co.
8	Welsh, J. R. (1981). Fundamentals of Plant Genetics and Breeding. John Wiley and Sons, New York.
9	Poehlman, J.M. (1987). Breeding Field Crops, 3rd Ed. AVI Publishing Co. Inc., Westport, Connecticut
10	Chopra, V.L. (2000). Plant Breeding: Theory and Practice 2nd Ed. Oxford & IBH, New Delhi.

THE NATIONAL COLLEGE AUTONOMOUS BASAVANAGUDI, BENGALURU=
PAPER – VII: CELL & MOLECULAR BIOLOGY (THEORY)

Programme Name	B.Sc. in BOTANY	Semester	V
Course Title	Cell & Molecular Biology (Theory)		
Course Code:	DSC-BOT-C13-T	No. of Credits	04
Contact hours	56 Hours	Duration of SEA/Exam	2 ½ hours
Formative Assessment Marks	40	Summative Assessment Marks	60

Course Pre-requisite (s):

Course Outcomes (Cos): After the successful completion of the course, the student will be able to:

CO1. Understanding of Cell metabolism, chemical composition, physiochemical and functional organization of organelle

CO2. Contemporary approaches in modern cell and molecular biology.

CO3. To study the organization of cell, cell organelles and biomolecules (i.e. protein, carbohydrate, lipid and nucleic acid)

Contents		
Unit-I	Plant Cell – ultrastructure and its components. Cell wall – types, composition and functions. Biological membranes – types, composition and transport (Plasma membrane, nuclear membrane and ER membrane). Plant Cell organelles – structure and function (Nucleus, Vacuole, Mitochondrion and Chloroplast) Cytoskeleton	14 hrs
Unit – II	Chromosome Biology – Types and structural organization of an eukaryotic chromosome (upto nucleosome model). Types of Chromosomes – normal, giant and supernumerary chromosomes. Cell cycle – phases of eukaryotic cell cycle, check points and role of protein kinases. Cell division – Mitosis and Meiosis and significance. Karyotype – types and significance, Programmed Cell Death (PCD)	14 hrs
Unit – III	Molecular Biology – Historical perspectives, DNA is the genetic material (Griffith's, Hershey and chase experiments). Nucleic Acids – DNA structure, composition, types and the mechanism of replication. A brief account of DNA repair mechanism, RNA – structure, composition and types. Central Dogma of Molecular Biology, Genetic code - salient features. Gene expression in prokaryotes (Transcription and Translation)	14 hrs
Unit – IV	Gene concept, Genomics and Proteomics. Gene regulation – Lac operon concept. Epigenetics – Gene editing – DNA methylation, sn/mi RNAs, Ribozymes. Genome organization in eukaryotes. Recombinant DNA Technology – a brief account. Introduction to Bioinformatics and its application.	14 hrs

Pedagogy: Teaching and learning, Seminar, Assignments, etc.

Formative Assessment for Theory	
Assessment Occasion/type	Marks
Attendance	10
Test	10
Assignments	10
Seminar	10
Total	40 Marks
<i>Formative Assessment as per NEP guidelines are compulsory</i>	

Course Title	Cell & Molecular Biology (Practical)	Practical Credits	02
Course Code:	DSC-BOT-C14-P	Contact Hours	52 Hours
Formative Assessment Marks	25 marks	Summative Assessment Marks	25 Marks
Practical Content			
1. Study of plant cell structure with the help of an epidermal peel mount of Onion/Rhoeo/Crinum. 2. Study of cell and its organelles with the help of electron micrographs. 3. Study of different stages of mitosis and meiosis (Onion/Rhoeo/Crinum) 4. Study of Karyotype using camera lucida / Chart. 5. Salivary gland chromosome 6. Isolation of cell organelle – Chloroplast. 7. Molecular Biology – Isolation of DNA by CTAB method (Cauliflower) 8. Estimation of RNA – by Orcinol method			

GENERAL PATTERN OF THEORY QUESTION PAPER

(60 marks for semester end Examination with 2 hrs duration) Part – A

1. Question number 1-06 carries 2 marks each. Answer any 05 questions: 10 marks

Part – B

2. Question number 07-11 carries 05 marks each. Answer any 04 questions: 20 marks

Part – C

3. Question number 12-15 carries 10 marks each. Answer any 03 questions: 30 marks

(Minimum 1 question from each unit and 10 marks question may be half sub-questions for 7+3 or 6+4 or 5+5 if necessary)

Total: 60 Marks Note: Proportionate weightage shall be given to each unit used on number of hours prescribed.

THE NATIONAL COLLEGE AUTONOMOUS BASAVANAGUDI, BENGALURU=
SCHEME OF PRACTICAL EXAMINATION
(distribution of marks): 25 marks for the Semester end examination
CELL & MOLECULAR BIOLOGY

Time=04 hrs

Marks=25

1. Preparation of squash/smear of material A, identify, Sketch and label any two stages with reasons
06 marks
2. Isolation of DNA/Estimation of RNA of material B 06 marks
3. Identify the slides C & D 04 marks (2M+2M)
4. Viva-voce 03 marks
5. Submission (Record + 4 slides) 06 marks (4M+2M)

Total 25 marks

General Instructions:

- Q1. Onion/Rhoeo/Crinum plant (A) Q2. Cauliflower head / RNA sample Q3. C - slide from mitosis or meiosis
D - Karyotype / Salivary gland chromosome Q4. Viva-voce
Q5. Submission (Record + 5 slides)

Note: Same Scheme may be used for IA (Formative Assessment) examination

References	
1	Cooper, G.M., Hausman, R.E. (2009). The Cell: A Molecular Approach, 5th edition. Washington, DC: ASM Press & Sunderland, Sinauer Associates, MA
2	Karp. G. (2010). Cell Biology, 6th edition. New Jersey, U.S.A.. John Wiley & Sons.
3	De Robertis, E. D. P. and De Robertis R. E. 2009. Cell and Molecular Biology, 8th edition. Lippincott Williams and Wilkins, Philadelphia.
4	Becker W. M., Kleinsmith LJ, and Bertni G. P. 2009. The World of the Cell. 7th edition. Pearson Benjamin Cummings Publishing, San Francisco.
5	Reven, F.H., Evert, R.F., Eichhorn, S.E. (1992). Biology of Plants. New York, NY: W.H. Freeman and Company
6	Alberts, B., Bray, D., Hopkin, K., Johnson, A. D., Lewis, J., Raff, M., Roberts, K., & Walter, P. (2013). Essential cell biology (4th ed.). Garland Publishing.
7	Raven, F.H., Evert, R. F., Eichhorn, S.E. (1992). Biology of Plants. New York, NY: W.H. Freeman and Co.
8	Verma, P. S. (2004). Cell Biology, Genetics, Molecular Biology: Evolution and Ecology, India: S. Chand Limited.

PAPER – VIII: PLANT PHYSIOLOGY AND PLANT BIOCHEMISTRY

Program Name	BSc/ BOTANY	Semester	VI
Course Title	PLANT PHYSIOLOGY AND PLANT BIOCHEMISTRY		
Course Code:	BOT C15-T	No. of Credits	04
Contact hours	4 Hours/ Week	Duration of Exam	2 1/2 hours
Formative Assessment Marks	40	Summative Assessment Marks	60

Course Pre-requisite(s):

Course Outcomes (COs): After the successful completion of the course, the student will be able to:

- Importance of water and the mechanism of transport.
- To understand biosynthesis and breakdown of biomolecules.
- Role of plant hormones in plant development and about secondary metabolites.
- Preliminary understanding of the basic functions and metabolism in a plant body.
- To understand the importance of nutrients in plant metabolism and crop yield.

Contents	56 Hrs
UNIT 1	14 Hrs
Plant water relations: Importance of Water as a solvent; Diffusion, osmosis, imbibition, osmotic potential, turgor pressure, wall pressure, water potential and its components. Mechanism of water absorption, Factors affecting water absorption. Transpiration. Types, Stomatal apparatus and mechanism of stomatal movement. Antitranspirants. Mechanism of ascent of sap: Vital and physical force theories. Phloem Transport: Transport of organic solutes. Path of transport, vein loading and unloading. Trans cellular hypothesis and mass flow hypothesis. Mineral nutrition: A brief account on micro and macro- nutrients.	
UNIT 2	14 Hrs
Photosynthesis: Photosynthetic Pigments (Chl a, b, xanthophylls, carotenes); Photosystem I and II, reaction centre, antenna molecules; Electron transport and mechanism of ATP synthesis; C3, C4 and CAM pathways of carbon fixation; Photorespiration. Respiration: Glycolysis, TCA cycle; Oxidative phosphorylation, Anaerobic respiration, Nitrogen metabolism: Biological nitrogen fixation; Nitrate and ammonia assimilation.	
UNIT 3	14 Hrs

Definition and classification of plant growth regulators- Hormones. Site of synthesis, and influence on plant growth development of individual group of hormone- Auxins, Gibberellins, cytokinins, ABA, ethylene . Synthetic growth regulators- classification, their effect on plant growth and development. Practical utility in agriculture and horticulture. Sensory Photobiology: Biological clocks, photoperiodism, function & structure of phytochrome, phototropin & cryptochrome. Senescence, Aging & Cell Death (PCD and Autophagosis). Plant Movements - tropisms	
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UNIT 4	14 Hrs
Carbohydrate metabolism : Cellulose and Starch – structure and function Enzymes - classification, kinetics and mechanism of action. Proteins : classification, structure - primary, secondary, tertiary and quaternary; Amino acids - a brief account Vitamins - classification, distribution, and function. Lipids : classification, structure and function of fatty acids. Secondary plant products : distribution of terpenes, phenolics and nitrogen containing compounds and their role in plants	

Pedagogy: Formative Assessment for Theory PAPER	
DSC C15 - T	
Assessment	Marks
Attendance	10
Test	10
Seminar	10
Assignment	10
TOTAL	40

GENERAL PATTERN OF THEORY QUESTION PAPER

[60 Marks for Semester end Examination with 2 1/2 Hours duration) PART – A

1. Question number 1 - 6 carries 2 marks each. Answer any 5 Questions [10 Marks]

PART -B

2. Question number 7 - 11 carries 5 marks each. Answer any 4 Questions [20 Marks]

PART - C

3. Question number 12-15 carries 10 marks each. Answer any 3 Questions [30 Marks]
(1 question from each unit and 10 marks question may have sub-questions for 7+3, or 6+4 or 5+5)

PRACTICAL - PEDAGOGY

Course Title	Plant Physiology and Biochemistry	Practical Credits	2
Course Code	BOT C16-P	Contact Hours	52 Hours (4 H/Wk)
Formative Assessment	25 Marks	Summative Assessment	25 Marks
Practical Content/ Experiments			

1. Conduct experiment to demonstrate the phenomenon of exosmosis and endosmosis.
2. Determine the osmotic pressure of the cell sap by plasmolytic method.
3. Demonstrate root pressure / transpiration pull in plants.
4. Compare the rate of transpiration from the two surfaces of leaf by cobalt chloride paper method
5. Demonstrate that oxygen is liberated in the process of photosynthesis.
6. Separation of photosynthetic pigments by paper chromatography and measure their R_f values.
7. Separate the chloroplast pigments by Arnon method
8. Isolate and identify the amino acids from a mixture using paper chromatography.
9. Study of Phototropism.
10. Qualitative tests for Starch, Protein, Reducing Sugars and Lipids.
11. Estimation of TAN (Titrable Acid Number) from Bryophyllum leaves /Aloe vera
12. Visit to Research Institute / Scientific Laboratory

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Formative Assessment for Practical	
Assessment	Marks
Attendance	5 Marks
Test	10 Marks
Research Institute visit & Visit report	5+5 Marks
Total	25 Marks

SCHEME OF PRACTICAL EXAMINATION
(Plant Physiology and Biochemistry)

Time – 4 Hours

Max. Marks = 25

- | | |
|--|-----|
| 1. Conduct major experiment 'A'
(Osmotic Potential / Paper chromatographic separation of Pigments) | 06 |
| 2. Comment on minor experiments B & C
(CoCl ₂ expt/ O ₂ evolution/ Root pressure/ Transpiration pull expts) | 3+3 |
| 3. Perform Biochemical Test of Sample 'D' | 04 |
| 4. Viva –Voce | 04 |
| 5. Practical record | 05 |

REFERENCES

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2. Jain V K, 2008. Fundamentals of Plant Physiology. S Chand and Co.
3. Kochhar P L, Krishnamoorthy H N. Plant Physiology. Atmaram and sons, Delhi.
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5. Malik CP, 2002. Plant Physiology. Kalyani publishers.
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7. Noggle GR, Fritz GJ, Introductory Plant Physiology. Prentice Hall of India.
8. Pandey SN, Sinha BK, 2006. Plant physiology. Vikas Publishing House, New Delhi.
9. Salisbury F B, Ross C W, 1992. Plant Physiology. CBS publishers and Distributors, New Delhi.
10. Sinha A K, 2004. Modern Plant Physiology. Narosa publishing House, New Delhi.
11. Srivastava H S, 2004. Plant physiology and Biochemistry. Rasthagi publications.
12. Verma V, 2007. Text Book of Plant Physiology. Ane Books Pvt. Ltd.
13. Buchanan BB, Gruissem W & Jones RL, 2004. Biochemistry & Molecular Biology of Plants. Ikt International Pvt. Ltd.
13. Hopkins WG & Huner NPA, 2009. Introduction to Plant Physiology, 4th Edition, John

Wiley

14. Taiz L, Zeiger E, Max Moller I & Murphy A 2018 Fundamentals of Plant Physiology, Sinaur Associates
15. Hans Walter-Heldt et al. 2011. Plant Biochemistry, Academic Press